Infinitely Many Integer Solutions of $y^2 + x^2y + xz^2 - y + 1 = 0$

Abstract

We give a completely explicit infinite family of integer solutions of

$$y^2 + x^2y + xz^2 - y + 1 = 0.$$

The construction fixes $x = -201$, converts the equation into an indefinite Pell-type norm equation, and iterates one integral norm-preserving transformation. The proof itself is a direct calculation with a binary quadratic form; a final section explains how the seed and the transformation were found.

1 The explicit family

Set

$$a = 515095, \qquad b = 36332, \qquad D = 201,$$

and define integer sequences $(W_n)_{n\geq 0}$ and $(Z_n)_{n\geq 0}$ by

$$\begin{aligned} W_0 &= 20640, & Z_0 &= 299, \\ W_{n+1} &= aW_n + DbZ_n, & Z_{n+1} &= bW_n + aZ_n. \end{aligned} \tag{1}$$

Thus $Db = 7302732$.

Theorem 1. *For every $n \geq 0$ and every $\sigma \in \{1, -1\}$,*

$$(x, y, z) = \bigl(-201,\, W_n - 20200,\, \sigma Z_n\bigr) \tag{2}$$

is an integer solution of

$$y^2 + x^2y + xz^2 - y + 1 = 0. \tag{3}$$

The triples in (2), indexed by (n, σ), are pairwise distinct. In particular, (3) has infinitely many integer solutions.

Proof. The two numerical identities on which the construction rests are

$$a^2 - Db^2 = 515095^2 - 201 \cdot 36332^2 = 1 \tag{4}$$

and

$$W_0^2 - DZ_0^2 = 20640^2 - 201 \cdot 299^2 = 20200^2 - 1. \tag{5}$$

For arbitrary integers W, Z, put

$$W' = aW + DbZ, \qquad Z' = bW + aZ.$$

A direct expansion, with the mixed terms cancelling, gives

$$\begin{aligned} (W')^2 - D(Z')^2 &= (aW + DbZ)^2 - D(bW + aZ)^2 \\ &= (a^2 - Db^2)(W^2 - DZ^2). \end{aligned}$$

By (4), the transformation therefore preserves the quadratic form $W^2 - DZ^2$. Applying this observation inductively to (1), and using (5), yields

$$W_n^2 - 201Z_n^2 = 20200^2 - 1 \qquad (n \geq 0). \tag{6}$$

Now take $x = -201$, $y = W_n - 20200$, and $z = \sigma Z_n$. Since

$$201^2 - 1 = 40400 = 2 \cdot 20200,$$

the left-hand side of (3) becomes

$$\begin{aligned}(W_n - 20200)^2 + (201^2 - 1)(W_n - 20200) - 201Z_n^2 + 1 \\ = W_n^2 - 201Z_n^2 - 20200^2 + 1 = 0\end{aligned}$$

by (6). Hence every triple in (2) is an integer solution.

It remains only to prove that infinitely many of these solutions are distinct. The initial values W_0, Z_0 are positive, and every coefficient in (1) is positive; hence $W_n, Z_n > 0$ for all n. Moreover,

$$W_{n+1} = aW_n + DbZ_n > W_n,$$

because $a > 1$ and $DbZ_n > 0$. Thus (W_n) is strictly increasing. Solutions arising from different indices have different y-coordinates, while the two signs at a fixed index are distinct because $Z_n > 0$. Hence all triples indexed by (n, σ) are pairwise distinct. This proves the theorem. □

The first two levels of the family are

$$(-201, 440, \pm 299)$$

and

$$(-201, 12815057468, \pm 903905885).$$

Equivalently, the recurrence is encoded by

$$W_n + Z_n\sqrt{201} = (20640 + 299\sqrt{201})(515095 + 36332\sqrt{201})^n. \tag{7}$$

2 How the construction was found

The point of the construction is not the large numerical coefficients; it is the sign of x and the norm structure that this sign exposes.

1. Forcing an indefinite Pell-type equation

Write $x = -A$, where A is a positive odd integer, and set

$$H = \frac{A^2 - 1}{2}.$$

Then (3) is

$$y^2 + (A^2 - 1)y - Az^2 + 1 = 0,$$

so completing the square gives the equivalent equation

$$(y + H)^2 - Az^2 = H^2 - 1. \tag{8}$$

This is indefinite. Once one solution is available, multiplication by a nontrivial solution of $u^2 - Av^2 = 1$ can preserve the right-hand side and generate further solutions. This explains why a negative value of x is natural. For comparison, fixing $x = A > 0$ would instead produce a positive-definite equation of the form

$$(y + H)^2 + Az^2 = H^2 - 1,$$

which has only finitely many integral possibilities for that fixed A.

2. Engineering one seed

For $x = -A$, the original expression can be grouped as

$$y^2 - y + 1 + A(Ay - z^2). \tag{9}$$

Consequently, it is enough to find integers y, r, z satisfying

$$y^2 - y + 1 = Ar^2, \qquad z^2 = Ay + r^2. \tag{10}$$

Indeed, substitution into (9) gives

$$Ar^2 + A\big(Ay - (Ay + r^2)\big) = 0.$$

A convenient successful choice is

$$A = 201, \qquad y = 440, \qquad r = 31, \qquad z = 299,$$

for which

$$440^2 - 440 + 1 = 201 \cdot 31^2, \qquad 299^2 = 201 \cdot 440 + 31^2. \tag{11}$$

Thus $(-201, 440, \pm 299)$ is the required seed. In the completed-square coordinate

$$W = y + \frac{201^2 - 1}{2} = y + 20200,$$

this seed is $(W, Z) = (20640, 299)$, which explains the initial values in (1).

3. Extracting the norm-one multiplier from the same seed

The first equation of (10) has a useful hidden form, since

$$4(y^2 - y + 1) = (2y - 1)^2 + 3.$$

For $U, V \in \mathbb{Q}$, write

$$N(U + V\sqrt{201}) = U^2 - 201V^2.$$

A direct expansion shows that $N(\alpha\beta) = N(\alpha)N(\beta)$. For the values in (11), the preceding identity becomes

$$879^2 - 201 \cdot 62^2 = -3. \tag{12}$$

Hence the quadratic expression

$$\eta = 879 + 62\sqrt{201}$$

has norm -3. Squaring and dividing by 3 gives the integral element

$$\begin{aligned}\varepsilon = \frac{\eta^2}{3} &= \frac{879^2 + 201 \cdot 62^2}{3} + \frac{2 \cdot 879 \cdot 62}{3}\sqrt{201} \\ &= 515095 + 36332\sqrt{201}.\end{aligned}$$

Its norm is $(-3)^2/3^2 = 1$, which is exactly identity (4). Multiplying $W + Z\sqrt{201}$ by ε gives

$$(aW + 201bZ) + (bW + aZ)\sqrt{201},$$

namely the recurrence (1). Thus the same arithmetic data that produce the seed also produce the norm-preserving operation that iterates it indefinitely.

Remark 1. Theorem 1 proves infinitude, which is one of the two alternatives in the problem. It does not assert that the displayed family exhausts all integer solutions of (3).